

FRAME FOR PERFORMANCE

SSS 1-3 | Water Productivity Garden

Frame and execute a research or engineering mission on water-balance and productivity analysis. Produce quantitative evidence that can support a design or management decision.

MISSION KIT water records, yield data, calculator

EVIDENCE water balance, ET, IWP or ETWP evidence

GM-441 - MISSION PATH

- DO** 1. Frame objective and performance metric. 2. Define system boundary, assumptions and constraints. 3. Frame with water records, yield data, calculator. 4. Quantify uncertainty and quality-control checks. 5. Model/compare scenarios where appropriate. 6. Validate, evaluate and recommend.
- PROVE** Provide a traceable dataset, calculation/model/diagram, validation or QC note, and decision statement for water balance, ET, IWP or ETWP evidence.
- SUCCESS** The method is reproducible, assumptions are explicit, uncertainty is addressed, and the recommendation is technically defensible.
- SAFETY** State data sources and assumptions. Follow teacher instructions and school garden safety rules.

AGRISHINE TASK

Propose one optimization or next-stage prototype/test for the Water Productivity Garden, including a measurable acceptance criterion.

WATER RESEARCH MISSION

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GM-442 - MISSION PATH

- DO** 1. Frame objective and performance metric. 2. Define system boundary, assumptions and constraints. 3. Design with water records, yield data, calculator. 4. Quantify uncertainty and quality-control checks. 5. Model/compare scenarios where appropriate. 6. Validate, evaluate and recommend.
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INSTRUMENT THE DESIGN

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GM-443 - MISSION PATH

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CALIBRATE FOR PERFORMANCE

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GM-444 - MISSION PATH

- DO** 1. Frame objective and performance metric. 2. Define system boundary, assumptions and constraints. 3. Calibrate with water records, yield data, calculator. 4. Quantify uncertainty and quality-control checks. 5. Model/compare scenarios where appropriate. 6. Validate, evaluate and recommend.
- PROVE** Provide a traceable dataset, calculation/model/diagram, validation or QC note, and decision statement for water balance, ET, IWP or ETWP evidence.
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GM-445 - MISSION PATH

- DO** 1. Frame objective and performance metric. 2. Define system boundary, assumptions and constraints. 3. Model with water records, yield data, calculator. 4. Quantify uncertainty and quality-control checks. 5. Model/compare scenarios where appropriate. 6. Validate, evaluate and recommend.
- PROVE** Provide a traceable dataset, calculation/model/diagram, validation or QC note, and decision statement for water balance, ET, IWP or ETWP evidence.
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VALIDATE THE DESIGN

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GM-446 - MISSION PATH

- DO** 1. Frame objective and performance metric. 2. Define system boundary, assumptions and constraints. 3. Validate with water records, yield data, calculator. 4. Quantify uncertainty and quality-control checks. 5. Model/compare scenarios where appropriate. 6. Validate, evaluate and recommend.
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EVALUATE FOR PERFORMANCE

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GM-447 - MISSION PATH

- DO** 1. Frame objective and performance metric. 2. Define system boundary, assumptions and constraints. 3. Evaluate with water records, yield data, calculator. 4. Quantify uncertainty and quality-control checks. 5. Model/compare scenarios where appropriate. 6. Validate, evaluate and recommend.
- PROVE** Provide a traceable dataset, calculation/model/diagram, validation or QC note, and decision statement for water balance, ET, IWP or ETWP evidence.
- SUCCESS** The method is reproducible, assumptions are explicit, uncertainty is addressed, and the recommendation is technically defensible.
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WATER RESEARCH MISSION

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GM-448 - MISSION PATH

- DO** 1. Frame objective and performance metric. 2. Define system boundary, assumptions and constraints. 3. Optimize with water records, yield data, calculator. 4. Quantify uncertainty and quality-control checks. 5. Model/compare scenarios where appropriate. 6. Validate, evaluate and recommend.
- PROVE** Provide a traceable dataset, calculation/model/diagram, validation or QC note, and decision statement for water balance, ET, IWP or ETWP evidence.
- SUCCESS** The method is reproducible, assumptions are explicit, uncertainty is addressed, and the recommendation is technically defensible.
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STRESS-TEST THE DESIGN

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EVIDENCE water balance, ET, IWP or ETWP evidence

- DO** 1. Frame objective and performance metric. 2. Define system boundary, assumptions and constraints. 3. Stress-test with water records, yield data, calculator. 4. Quantify uncertainty and quality-control checks. 5. Model/compare scenarios where appropriate. 6. Validate, evaluate and recommend.
- PROVE** Provide a traceable dataset, calculation/model/diagram, validation or QC note, and decision statement for water balance, ET, IWP or ETWP evidence.
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RECOMMEND FOR PERFORMANCE

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MISSION KIT water records, yield data, calculator

EVIDENCE water balance, ET, IWP or ETWP evidence

GM-450 - MISSION PATH

- DO** 1. Frame objective and performance metric. 2. Define system boundary, assumptions and constraints. 3. Recommend with water records, yield data, calculator. 4. Quantify uncertainty and quality-control checks. 5. Model/compare scenarios where appropriate. 6. Validate, evaluate and recommend.
- PROVE** Provide a traceable dataset, calculation/model/diagram, validation or QC note, and decision statement for water balance, ET, IWP or ETWP evidence.
- SUCCESS** The method is reproducible, assumptions are explicit, uncertainty is addressed, and the recommendation is technically defensible.
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